PhonePe

PHONEPE DATA VISUALIZATION DASHBOARD

Business Intelligence and Digital Payment Analytics Using Power BI

PROJECT REPORT

Submitted in partial fulfillment of the requirements for the completion of the

DATA ANALYTICS / BUSINESS INTELLIGENCE PROJECT

PREPARED BY

AYUSH DASH

TOOLS & TECHNOLOGIES USED

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Modeling
- Data Visualization

PROJECT FEATURES

- Executive Overview Dashboard
- Transaction Analysis Dashboard
- User Analysis Dashboard
- Top Insights Dashboard
- Interactive KPI Monitoring
- Geographic Analysis
- Business Intelligence Reporting

PhonePe Data Visualization Dashboard: Transforming Digital Payment Data into Actionable Business Insights through Interactive Power BI Reporting.

Table of Contents

1. Executive Summary
2. Introduction
3. Business Problem Statement
4. Project Objectives
5. Data Source Overview
6. Dataset Description
7. Data Preparation Using Power Query
8. Data Modeling
9. DAX Measures and KPI Development
10. Dashboard Design and Navigation
11. Dashboard Analysis
    - Executive Overview
    - Transaction Analysis
    - User Analysis
    - Top Insights Analysis
12. Key Findings and Insights
13. Business Recommendations
14. Challenges and Solutions
15. Project Limitations
16. Future Enhancements
17. Conclusion
18. Appendix

1. Executive Summary

The PhonePe Data Visualization Dashboard is a Business Intelligence solution developed using Power BI to analyze India's digital payment ecosystem. The project transforms raw PhonePe Pulse datasets into interactive dashboards that provide insights into transaction trends, user growth, mobile brand adoption, and regional performance.

The solution enables stakeholders to monitor digital payment activity across states and districts, identify high-performing regions, and support data-driven decision-making. Through data modeling, DAX calculations, and interactive visualizations, the dashboard provides a comprehensive view of transaction behavior and user engagement across India.

2. Introduction

The rapid growth of digital payments in India has generated vast amounts of transaction and user data. Analyzing this information is critical for understanding customer behavior, monitoring platform performance, and identifying market opportunities.

This project leverages PhonePe Pulse data to create an interactive Power BI dashboard that visualizes payment activity, user adoption trends, and geographic performance across multiple regions.

The dashboard is designed to provide actionable insights through intuitive visualizations and dynamic filtering capabilities.

3. Business Problem Statement

Organizations operating in the digital payments sector require efficient tools to monitor transaction activity and user growth across multiple geographic regions.

However, raw datasets often contain millions of records that are difficult to analyze without a centralized reporting system.

The primary business challenges include:

- Monitoring transaction growth trends.
- Identifying top-performing states and districts.
- Understanding user adoption behavior.
- Comparing payment categories.
- Supporting strategic business decisions.

To address these challenges, an interactive Power BI dashboard was developed to consolidate and visualize key business metrics.

4. Project Objectives

The primary objectives of this project were:

Transaction Analysis

- Analyze transaction volume and transaction value.
- Evaluate transaction category distribution.
- Track quarterly transaction growth.

User Analysis

- Monitor registered user growth.
- Analyze mobile brand market share.
- Evaluate user adoption trends.

Geographic Analysis

- Compare state-level performance.
- Identify top-performing districts.
- Visualize regional payment activity.

Business Intelligence

- Create executive-level KPIs.
- Enable interactive exploration of data.
- Support strategic decision-making.

5. Data Source Overview

The project utilizes PhonePe Pulse datasets, which provide publicly available information regarding digital payment transactions and user statistics across India.

The datasets include:

- Transaction data
- User data
- Geographic data
- Mobile brand data

These datasets were imported directly into Power BI for analysis and visualization.

6. Dataset Description

Dataset	Description
Aggregated Transaction	State-level transaction statistics
Aggregated User	State-level user statistics
Map Transaction	Geographic transaction data
Map User	Geographic user data
Top Transaction	Top-performing districts by transaction value
Top User	Top-performing districts by user count

Key Attributes

- State
- District
- Year
- Quarter
- Transaction Type
- Transaction Count
- Transaction Amount
- Registered Users
- App Opens
- Mobile Brand

7. Data Preparation Using Power Query

Several transformation steps were performed to ensure data quality and consistency.

Activities Performed

- Imported multiple CSV files.
- Validated data types.
- Renamed columns.
- Removed unnecessary fields.
- Standardized geographic names.
- Handled missing values.
- Optimized dataset structure.

Outcome

A clean and analysis-ready dataset suitable for advanced reporting and visualization.

8. Data Modeling

A relational data model was created within Power BI to establish relationships between transaction and user datasets.

Model Components

Fact Tables

- Aggregated Transaction
- Aggregated User

Supporting Tables

- Top Transaction
- Top User
- Map Transaction
- Map User

Benefits

- Faster calculations
- Improved report performance
- Consistent filtering across dashboards

9. DAX Measures and KPI Development

Several DAX measures were created to support business analysis.

Transaction KPIs

```
Total Transaction Amount =

SUM(aggregated_transaction[Transaction_amount])
```

```
Total Transactions =

SUM(aggregated_transaction[Transaction_count])
```

```
Average Transaction Value =

DIVIDE(

[Total Transaction Amount],

[Total Transactions]

)
```

User KPIs

```
Total Registered Users =

SUM(aggregated_user[RegisteredUsers])
```

```
Total App Opens =

SUM(aggregated_user[AppOpens])
```

Ranking KPIs

```
State Rank =

RANKX(

ALL(State),

[Total Transaction Amount]

)
```

```
District Rank =

RANKX(

ALL(District),

[Top Transaction Amount]

)
```

10. Dashboard Design and Navigation

The dashboard consists of four interconnected pages:

Executive Overview

Provides a high-level summary of platform performance.

Transaction Analysis

Focuses on transaction value, volume, and payment categories.

User Analysis

Focuses on registered users and mobile brand performance.

Top Insights

Highlights key rankings and comparative analyses.

Interactive navigation buttons and slicers enable seamless movement across dashboard pages.

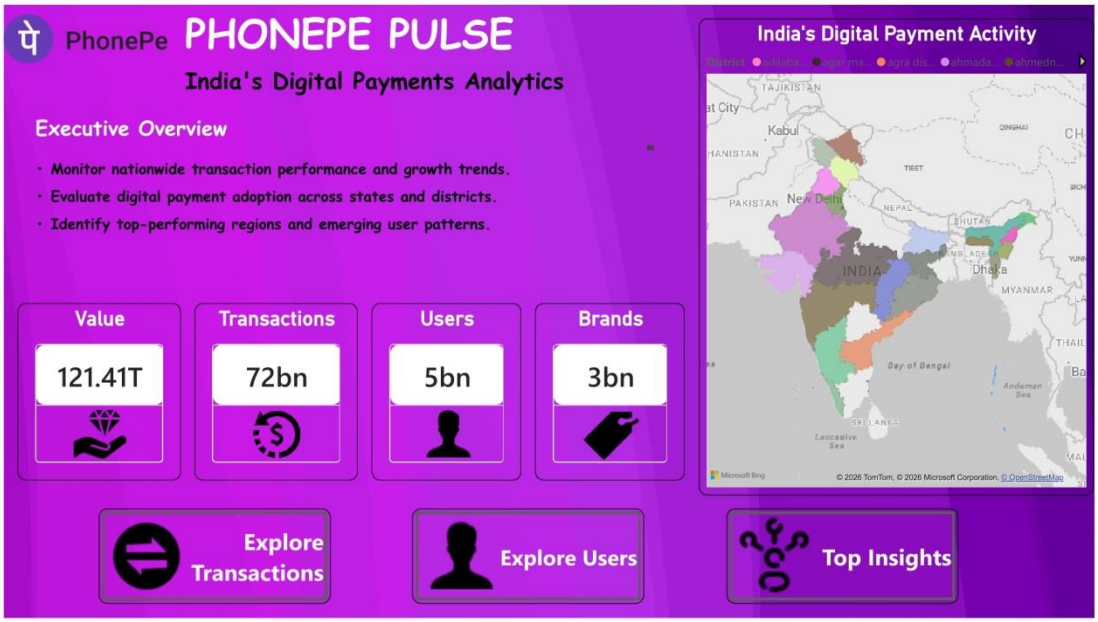
11. Dashboard Analysis

Executive Overview

The dashboard displays overall platform metrics including:

- Transaction Value: 121.41 Trillion
- Transactions: 72 Billion
- Users: 5 Billion
- Brands: 3 Billion

These metrics provide a snapshot of India's digital payment ecosystem.

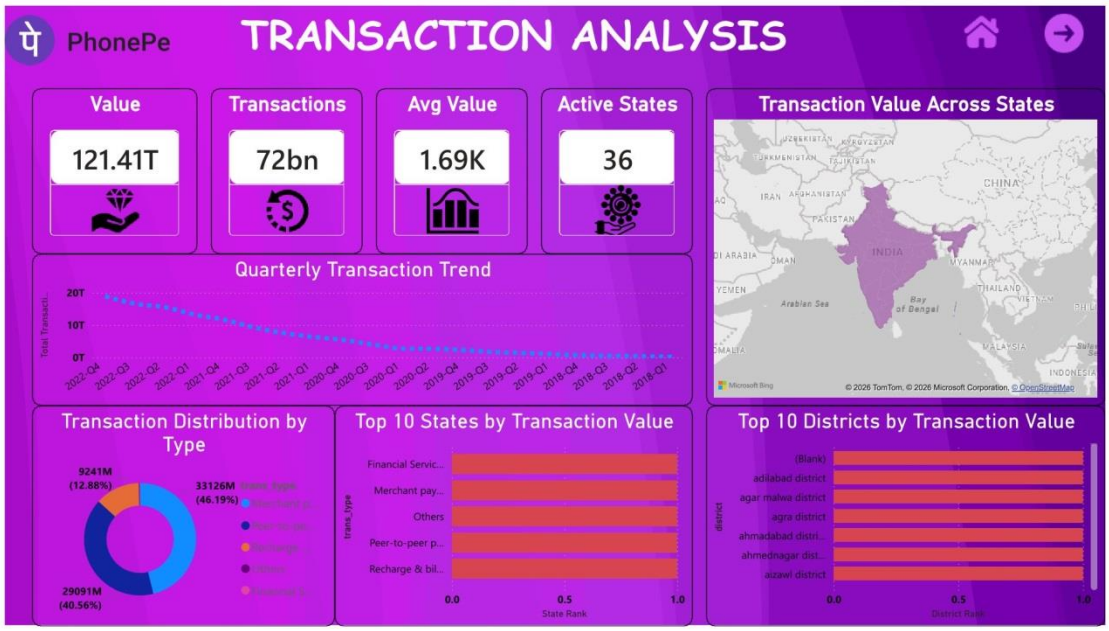


Transaction Analysis

The transaction dashboard highlights:

- Quarterly transaction trends.
- Transaction type distribution.
- State-wise transaction performance.
- Top districts by transaction value.

Merchant payments represent the largest transaction category, followed by peer-to-peer payments.

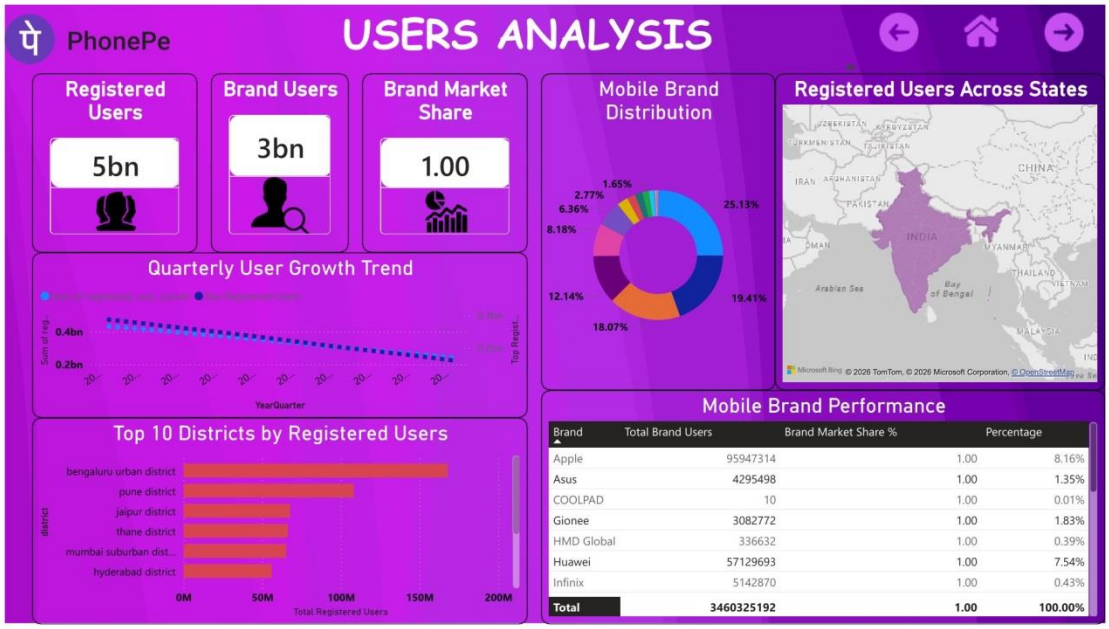


User Analysis

The user dashboard focuses on:

- Registered users across states.
- Mobile brand market share.
- User growth trends.
- Top districts by registered users.

The analysis reveals strong user concentration in major urban districts such as Bengaluru Urban and Pune.

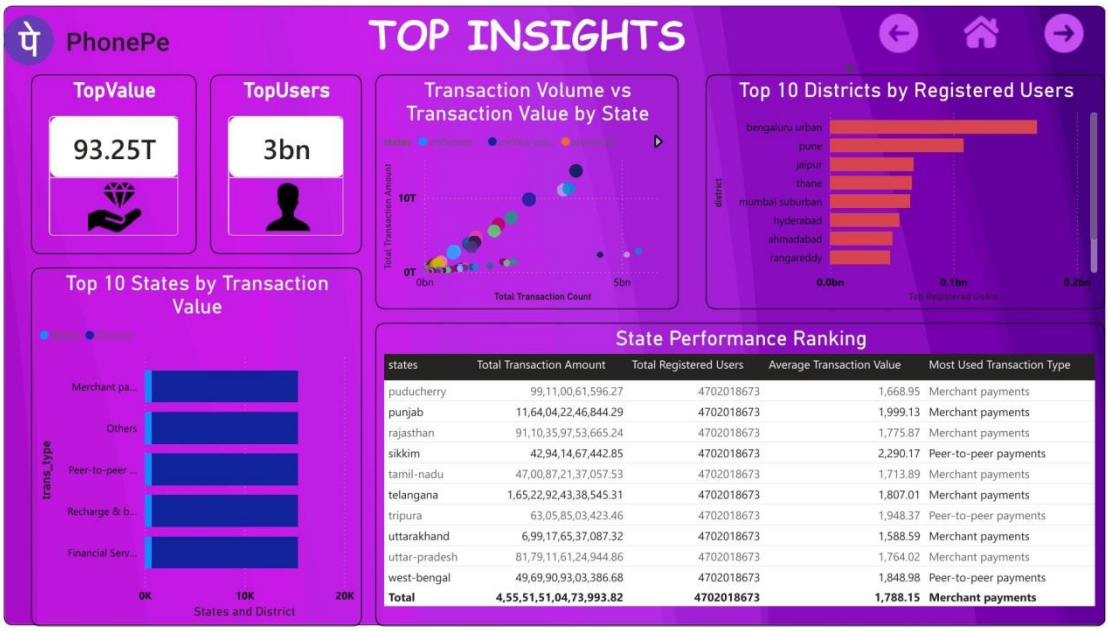


Top Insights Dashboard

This dashboard combines transaction and user insights to identify:

- Top-performing states.
- High-value districts.
- Transaction volume versus value relationships.
- State performance rankings.

The analysis provides management with a concise view of market leaders and growth opportunities.



12. Key Findings and Insights

Transaction Insights

- Total transaction value exceeded ₹121 Trillion.
- Merchant payments contributed the largest share of transactions.
- Transaction activity increased consistently across quarters.

User Insights

- Total registered users exceeded 5 Billion.
- Bengaluru Urban district recorded the highest user adoption.
- Mobile brand usage is concentrated among a few leading manufacturers.

Geographic Insights

- Significant regional variation exists across states.
- High-performing districts contribute disproportionately to transaction value and user growth.

13. Business Recommendations

Regional Expansion

Focus marketing efforts on underperforming states with growth potential.

User Acquisition

Target districts with lower user penetration rates.

Customer Engagement

Implement loyalty programs and incentives to increase app activity.

Strategic Partnerships

Use mobile brand analytics to identify partnership opportunities with device manufacturers.

Performance Monitoring

Regularly monitor transaction growth trends to identify emerging markets.

14. Challenges and Solutions

Challenge	Solution
Large dataset size	Optimized Power Query transformations
Data inconsistencies	Standardized field names
Complex KPI calculations	Developed DAX measures
Dashboard performance	Implemented efficient data modeling
Geographic visualization issues	Used Power BI mapping features

15. Project Limitations

- Historical data limited to available PhonePe Pulse records.
- No real-time data integration.
- Limited demographic information.
- No predictive analytics implementation.

16. Future Enhancements

- Real-time dashboard refresh.
- Predictive transaction forecasting.
- Customer segmentation analysis.
- AI-driven recommendations.
- Fraud detection analytics.
- Mobile-optimized reporting.

17. Conclusion

The PhonePe Data Visualization Dashboard successfully transformed raw transaction and user data into meaningful business insights using Power BI.

The project demonstrates the practical application of data modeling, DAX, and visualization techniques to solve real-world business problems. The dashboard enables stakeholders to monitor digital payment performance, understand user behavior, and make informed strategic decisions.

18. Appendix

Tools Used

- Power BI Desktop
- Power Query
- DAX

Dashboard Pages

1. Executive Overview
2. Transaction Analysis
3. User Analysis
4. Top Insights

KPIs Created

- Total Transaction Amount
- Total Transactions
- Average Transaction Value
- Total Registered Users
- State Rank
- District Rank
- Top Transaction Amount
- Top Registered Users